

Donate.lk: A smart donation handling system

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Abstract— Sri Lanka is a small island in the Indian Ocean which is in the path of two monsoons. Therefore, the country is mostly affected by weather related hazards. Floods are one of the most common hazards that the country faces due to monsoonal rain or effects of low pressure systems. Droughts due to failure of monsoonal rain are another common hazard experienced in Sri Lanka. Moreover, Sri Lanka is also prone to hazards such as landslides, lightning strikes and coastal erosion. In 2004, almost two-thirds of the Sri Lankan coast was affected by the Indian Ocean tsunami. State, Non-governmental and Media organizations in Sri Lanka initiate much needed relief to people affected by floods and landslides across the country by organizing donation campaigns. People from all around the country help victims through those campaigns. While delivering those rations, there have been many conflicts related to identifying most needed people and what kind of rations they needed. During such disasters, manual labor is hard to find. At present there is no proper system to manage those donation campaigns to verify equal distribution of donations to victims. People handle those with difficulty by making groups through social media which is a temporary solution. All of the above matters indicate that, a country like ours must have an automated system to manage donations/rations. Also, by using global servers, an optimum route will be created to make sure that the rations are delivered and provided to the people in need. Therefore, as a solution, this paper proposes a system that manages donations and delivers them to the people in need with fewer burdens.

Keywords— *Keywords-disasters, flood, donations/ration, manual labor, automate, Geographic Information System*

I. INTRODUCTION

Natural disasters are common all over the world and are based on geophysical, hydrological, climatological or biological incidents. In the past history, Sri Lanka faced lot of natural disasters due to bad weather conditions starting from floods to landslides. Those disasters cause major damages such as loss of human lives and loss of properties. During the

last years, there has been an increase in the reports of natural disasters.

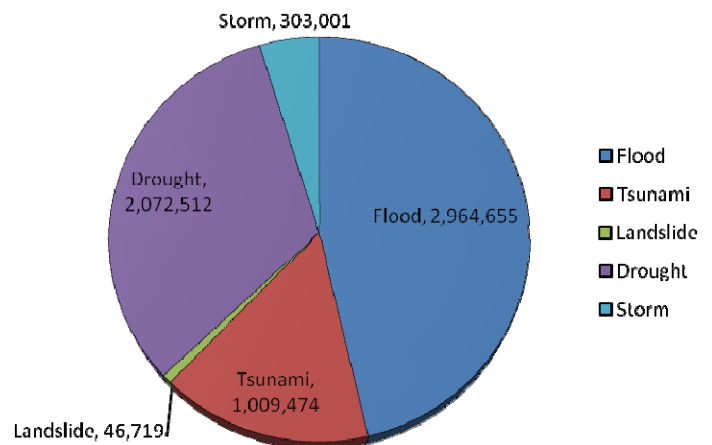


Fig. 1: People affected by different disasters in Sri Lanka (1974-2004) [14]

According to the above figure (figure 01), many people are affected by natural disasters and most of them from floods. In 2017, provinces such as western province, southern province, and Sabaragamuwa province were highly affected by floods and earth slips. Around 557,505 People belonging to 151,392 families were affected in 15 Districts due to flooding and landslides [1]. During a period of a disaster, people face so many difficulties in carrying out their day to day activities. The common problems are to find food, clean water, medicine and other sanitary items, which they need for a healthy life. In Sri Lanka, many people and organizations are willing to support these people and there are volunteers who are willing to collect the items donated by people and organizations and distribute them among victims. Most of the time, the distribution teams have to face lot of problems in issues such

as methods of delivery, most needed items and quantities and whether they distribute them to true victims. While investigating these situations further, researchers identified numerous misconducts in this process of distributing donations. Those misconducts affected the victims in higher scale, since in most scenarios, the rations were not distributed equally and for those who needed most. As a result, lots of victims who were affected by the floods fell into grave depression.

This product was developed after investigating current disaster management system in Sri Lanka and in several other countries which get affected by natural disasters frequently. According to the initial investigation, there are lot of problems and requirements to be addressed to ensure fair distribution of donations among victims. By looking into those conflicts and requirements, this proposed system will make sure that the people who are affected will get the necessities and that their needs are fulfilled.

Following table represents the existing products and features that are available with regard to disaster management and donation handling. Each and every product has its own unique features, but what one product lacks is provided by another product.

Table1: Existing disaster supporting systems

Product	Functions
Sahana Foundation [2]	Represents physical features on the ground. Send warnings and situation-awareness.
AIT System [3]	Provide a wide verity of associated notification services, systems and solutions.
ICRC's Family Links website [4]	Help people separated by conflict.

When comparing Sahana foundation with the other two supporting services, they host their own map API's similar to Google Map API's [5]. Open street maps are customized by illustrating routes that help volunteers to reach affected areas easily [6]. This feature supports people who are affected by floods or other natural disasters efficiently and it ensures that affected parties can get help with fewer burdens. Other two services lack above features which are very important in a disaster management system [7].

The solution proposed through this research combines the necessary features such as managing donations, supporting donation handling teams and providing volunteers/teams to reach the people in need in an optimum and the safest way possible. The proposed donation handling system is a unique product that is very much needed for people who face those disasters and for the people who are willing to help the people in need.

II. METHODOLOGY

The proposed system consists of two components, mobile application and web application. The mobile application is compatible with all android versions up to version 6 and the web application is developed using AngularJS.

A. Web Application

The web application is developed to manage users of the donation teams. The system administrator can add users for donation teams and assign appropriate privileges to them. The assigned users then can access functionalities in the mobile application.

B. Mobile Application

Mobile application consists of five sub functions which collaborate together to work as a system. Figure 02 shows how the main functionalities of the proposed system work in an active session.

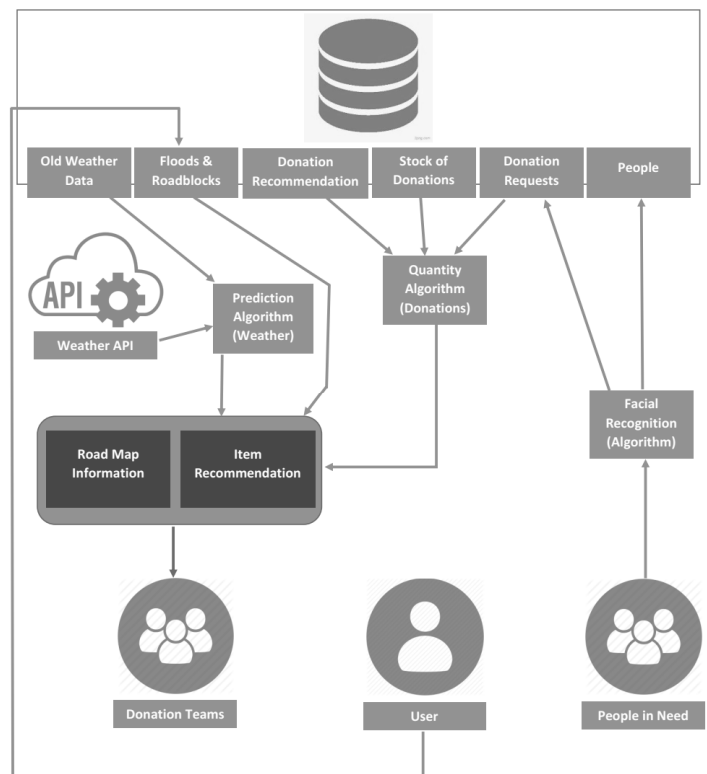


Fig.2: Overall system diagram of the proposed solution

C. Profiling of old Weather data

As represented in the Figure 02, the database used in the proposed system consists of several sub databases. Old weather data is collected for data mining purposes. Floods and Roadblocks are updated frequently with the updates of

weather reports and also from the information received from the victims. Stock of Donations is fed into the database and donation requests are managed in a separate database. This makes request retrieval more flexible. People and Donation Recommendations are managed according to the information gathered from the database including donated item stocks so that the profiling would be possible and allocations can be made easily.

D. Facial Identification using the camera

This function uses the front facing camera of a smart phone to capture faces of the victims. Microsoft Cognitive Services (Face API) will be used to identify and record the faces of the people/victims in an array for future accesses [8]. Subscription keys in the database will support to verify facial records while using Face API and to track down faces of the given. Every facial record will create an entry so that there will not be any redundancies of facial records of those people in need. Otherwise the probability of succession in identifying people would fail. Donate.lk team assures that personal information of individuals is protected and it promises not to give that information to any third-party organization.

```
const setFaceData = (obj, res) => {
  if (obj.photoUrl) {
    const api_url =
'https://eastasia.api.cognitive.microsoft.com/face/v1.0'
    const api_params =
'/detect?returnFaceId=true&returnFaceLandmarks=false&returnFaceAttributes=age,gender'
    const headers = {
      'Accept': 'application/json',
      'Content-type': 'application/json',
      'Ocp-Apim-Subscription-Key':
'417def654dd94563b0146f197d06cc18'
    }
    const body = {url:obj.photoUrl}

    return fetch(api_url+api_params, {
      method:'POST',
      headers: headers,
      body: JSON.stringify(body)
    }).then(async _res => {
      console.log(_res)
      const json = await _res.json()
      obj.faceData = json
      obj.faceCount = json && json.length
      addReq(obj, res)
    }).catch(err => console.error(err))
  } else {
    addReq(obj, res)
  }
}
```

Fig.3: Get the face analysis from API

E. Donation Classification

Donations that are received from the individual donors or support organizations will be categorized according to the details of the victims. A supervised data classification algorithm, Classification of Task data accessing, and arithmetic computing procedures will be used for data mining purposes. [9] Data mining is used to understand the types of people who are fit to receive donations and to clarify which person would receive what kind of donations from the available stocks. In this section, data mining is mainly used for profiling donation stocks and categorizing those victimized people. The amount of each type of items to be taken from stock for a specific donation request is calculated based on the request of the people in the given location. For an example, if there are 2 pregnant women, 3 men, 4 kids and 5 women, then the preconfigured profiles will be allocated as per the requests that are received from those above said people. Mainly it implies that the number of victims will receive one of each item under that specified profile or the category of the person once the stocks are available to share among the needy.

```
app.post('/donations/bulk', (req,res) => {
  const donationsList = req.body
  console.log('donationsList', donationsList)

  donationsList.forEach(donation => {
    colls.donations.add(donation).then(err => {
      colls.stocks.getByCriteria({'__name__':'master'},
      (err, stock) => {
        console.log(donation, stock)
        stock[donation.category] += (+donation.amount)
        console.log(stock)
        colls.stocks.setFields({'__name__':'master'}, stock)
      })
      res.json()
    }) }) })
```

Fig.4: Get the donation details

F. Road Map Information

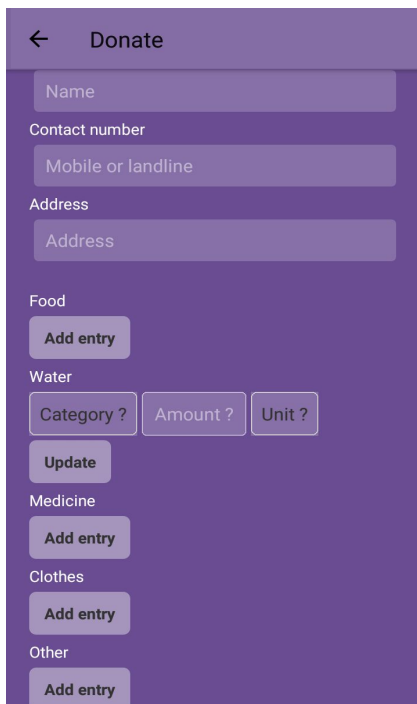
Google map API [10] is used to create a virtual map for the donation team. That map will be compared and analyzed once the victim submits his/her GPS locations by using the application. After receiving victim's location, the application will suggest an optimum route for the donation team to reach that person in need.

G. Weather prediction using Dark API

The surveys collected under 'weather' category for the past year is not sufficient to predict the possible weather for certain given date in the upcoming months. Compared to those surveys, even online weather predictions are not that accurate for specific areas in Sri Lanka. For an example, if we were to look in-to the rainfall for the coming two weeks, one might have to refer to those above said survey data and also online weather API's altogether [11]. Even though we refer to two sources, the accuracy rate will be between 60-70% even after a fine tuned algorithm is used for prediction. As per existing algorithms, 60-70% is the most available level of predictability, a prediction algorithm can have. Therefore, the proposed system will use the Dark Sky API [12] to get the forecasts by analyzing historical weather data later to implement on other functionalities of the proposed systems.

III. DISCUSSION

The proposed system is mainly implemented as a mobile application. Therefore, a Huawei P9Lite with GPS and LTE enabled smartphone was used for testing purposes. Android 6.0, [13] a Linux based mobile operating system has been used for algorithm and image processing to execute. Java has been used as the algorithm execution language under this demonstration. A storage size of 120MB was used for installation of this application. Initialization of the application takes up to 12 seconds. Google map loading time depends on the free memory available in the smartphone itself. Communication between the application and the weather API's takes up to 3-4 seconds.



The screenshot shows a mobile application interface for donating items. It features a purple header with a back arrow and the title 'Donate'. Below the header, there are input fields for 'Name', 'Contact number' (with a sub-label 'Mobile or landline'), and 'Address'. There are three main categories for donation: 'Food', 'Water', and 'Medicine'. Each category has an 'Add entry' button. The 'Water' category also includes sub-fields for 'Category?', 'Amount?', and 'Unit?'. There is an 'Update' button below the 'Water' section. At the bottom, there are two more categories: 'Clothes' and 'Other', each with an 'Add entry' button.

Fig.5: Donating application for item donation

According to the above figure (Figure 05), a person or a company that is willing to support by donating to the cause can fill out the form including what to donate. This criterion is only eligible for the system administrator. The confirmation will happen when the stocks of donations are received.

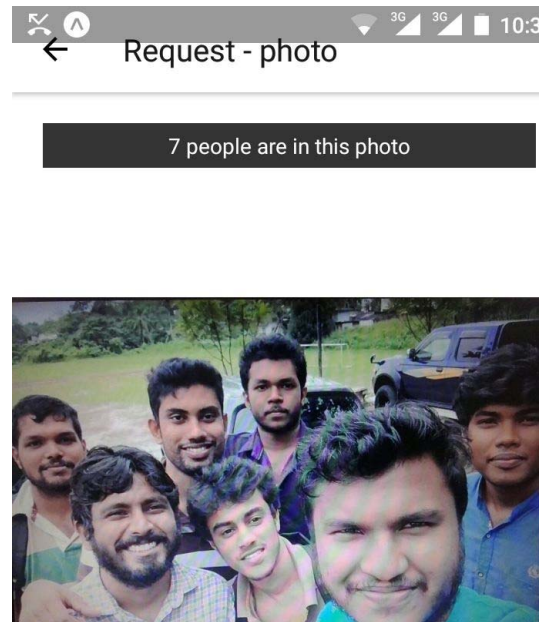


Fig.6: Get victim count

A victim can introduce his/her group of people in need in the area. One request can handle up to 15 people. A request can be added with a specific location and the user can add a photograph of those affected by the flood and submit it to the system.

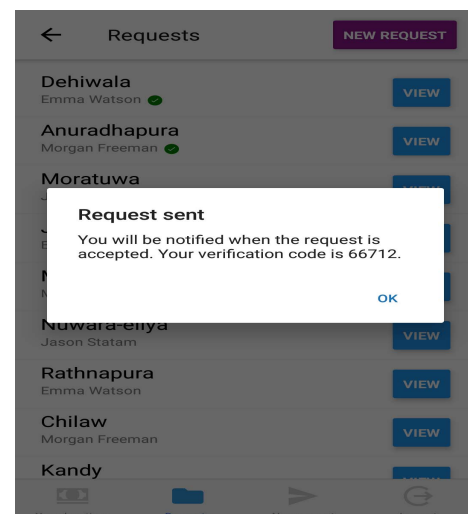


Fig.7: Conformation alert for a request

The requester will receive a verification code after the request is made. Once the donation team arrives at the requested location, the donation team leader can confirm that above said verification code with the request remark in their application.

When considering the facial recognition (Facial API), the system might not fully understand the victim faces. Currently this system uses the free account of the Azure Facial API which is limited to 30 000 transactions per month with 20 transactions per minute, and thus. The need to use the Face API Standard package to make this application as an official product.

Even a person with little knowledge on mobile applications can understand about the Application Programming Interfaces (API's) and their data retrieval timeouts. That is due to interrupted internet connections and incompatible operating system platforms.

Inserting incorrect data to the dataset will result in inaccurate analysis on the stock donations and profiling. Therefore, it is important to filter out the datasets before inserting them into the data mining algorithms. Similarly, it is important to input online weather forecast datasets to use in prediction algorithms. Following diagram displays the average monthly wind speed between 2013-2017. It is useful for calculating accurate weather results.

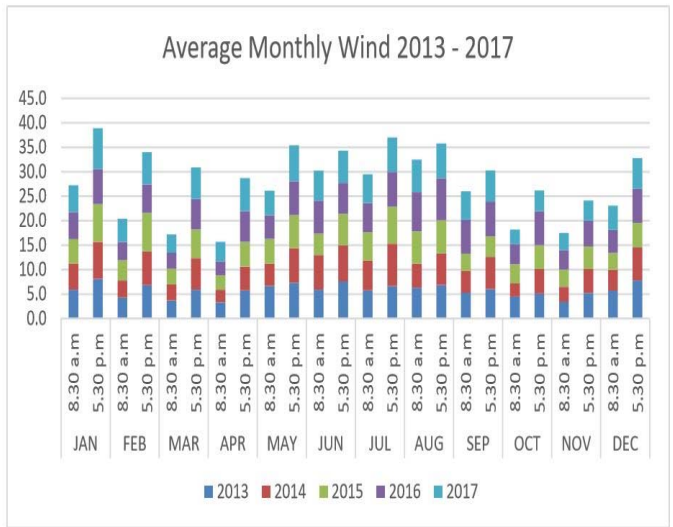


Fig.8: Average wind speed

IV. CONCLUSION AND FUTURE WORK

Donate.lk is a mobile application developed to enhance the donation handling process in Sri Lanka at a time of a disaster using latest technologies. The main objective of this research is to provide a better solution for donation handling. The system consists of a web application and the donation handling process mainly functions through a mobile application. This system identifies similar faces using an image processing technique which is not available in any other disaster management applications. Therefore, those allocations will be made by analyzing the desired donations and available donation teams. Stocks of donations and Profiling of those stocks are mandatory even before setting out to reach the people in need.

This system will expand using mobile ad-hoc network technology. It may be helpful when mobile communications are down during disasters. The proposed solution can be enhanced by several methods. One futuristic idea would be to expand the proposed application as an island wide system and to improve the system to handle multiple requests without any delays in analyzing and allocating resources. For further clarifications, the proposed system can be improved to identify identical twins of a family in crisis.

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